BLK II - UNIT 2: Basic Forecast Techniques
a. Forecasting Tools

The goal of this objective will be to show you have the capability to identify the intensity and movement of weather features based on meteorological reasoning. To successfully demonstrate competency on this topic, you will be given information through lecture, a series of questions and tasks designed to display your ability to correctly predict the future state of the atmosphere using the tools discussed throughout this objective. These foundational forecasting concepts will be measured with a Progress Check (PC) at the end of the objective and must be passed to continue onto the next objective. Follow your instructor's guidance as they provide information and instruction for completing all requirements necessary to pass this objective.

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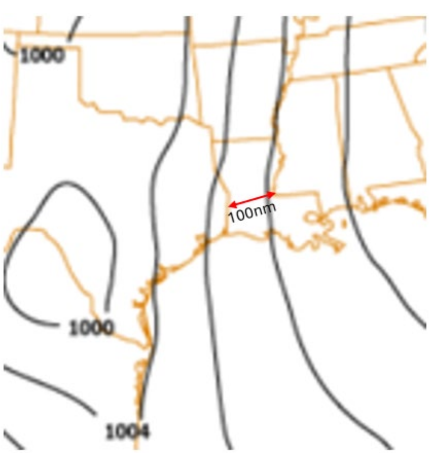
Prognosis Rules & Forecast Techniques

Prognosis rules & forecast techniques refer to the ability to use your knowledge of physics and dynamics to predict the formation, dissipation, intensity changes, and movement of ~~weather~~ the weather features you learned how to analyze previously. These rules and techniques can be useful if you are working with limited data but still need to inform decision makers on potential hazards. By applying some physics and dynamics, as well as previously observed patterns, you can still have an informed forecast that can impact your mission.

A well-known forecast technique is, "The time in seconds between seeing lightning strikes and hearing thunder will tell you the distance of the storm from your location in miles". This is a rough estimate due to the differences in the speed of light vs speed of sound, but if that is the only data you have, it is better than just guessing. Another useful forecast technique is, "one contour/isobar per 100 miles produces wind speeds near 10 knots". So, if there were three contours in the 100-mile area, the increasing gradient results in wind speeds near 30 knots. These forecast techniques should be used as a general starting point to have an idea of what can be expected. These should not be used on their own unless no other tools are available to you.



Anticyclogenesis
Anticyclogenesis describes the formation and building of anticyclonically oriented features. Anticyclogenesis can be forecasted by analyzing:

Damper Effect: Convergence aloft, or divergence on the surface would result in downward vertical motion. This would cause an increase in surface pressure resulting in anticyclogenesis.

Warm / Cold Air Advection: Increasing warm air advection upstream and/or cold air advection downstream of an anticyclonically oriented feature would result in building of the anticyclonic rotation resulting in anticyclogenesis.

Diabatic Effects: Surface based cold core features such as a high-pressure center moving over a relatively colder surface would cause the cold air within the system to increase resulting in anticyclogenesis.

Diurnal Effects: Surface based cold core features during night-time cooling would build in intensity as the atmosphere affecting the feature would become relatively colder than during the hottest part of the day resulting in a more stable atmosphere and anticyclogenesis.

Anticyclolysis
Anticyclolysis describes the dissipation and weakening of anticyclonically oriented features. Anticyclolysis can be forecasted by analyzing:

Chimney Effect: Divergence aloft, or convergence at the surface would result in upward vertical motion. This would cause surface pressure to decrease resulting in anticyclolysis.

Warm / Cold Air Advection: Increasing cold air advection upstream and/or warm air advection downstream of an anticyclonically oriented feature would result in weakening the anticyclonic rotation resulting in anticyclolysis.

Diabatic Effects: Surface based cold core features such as a high-pressure center moving over a relatively warmer surface would cause the cold air within the system to begin absorbing the heat and decreasing surface pressure resulting in anticyclolysis.

Diurnal Effects: Surface based cold core features during daytime heating would decrease in intensity as the atmosphere affecting the feature would become relatively warmer than during nigh-time cooling resulting in a decrease in pressure and anticyclolysis.

Cyclogenesis
Cyclogenesis describes the formation and deepening of cyclonically oriented features. Cyclogenesis can be forecasted by analyzing:

Chimney Effect: Divergence aloft, or convergence at the surface would result in upward vertical motion. This would cause surface pressure to decrease, resulting in cyclogenesis.

Warm / Cold Air Advection: Increasing cold air advection upstream and/or warm air advection downstream of a cyclonically oriented feature would result in deepening the cyclonic flow resulting in cyclogenesis.

Diabatic Effects: Surface based warm core features such as a low-pressure center moving over a relatively warmer surface would cause the colder air within the system to begin absorbing the heat and decreasing surface pressure resulting in cyclogenesis.

Diurnal Effects: Surface based warm core features during daytime heating would increase in intensity and deepen as the atmosphere affecting the feature would become relatively warmer than during nigh-time cooling resulting in a decrease in pressure and cyclogenesis.

Cyclolysis
Cyclolysis describes the dissipation and filling of cyclonically oriented features.
Cyclolysis can be forecasted by analyzing:

Damper Effect: Convergence aloft, or divergence on the surface would result in downward vertical motion. This would cause an increase in surface pressure resulting in cyclolysis.

Warm / Cold Air Advection: Increasing warm air advection upstream and/or cold air advection downstream of a cyclonically oriented feature would result in the filling of the cyclonic flow and cyclolysis.

Diabatic Effects: Surface based warm core features such as a low-pressure center moving over a relatively colder surface would cause the pressure center to release heat and increasing pressure resulting in cyclolysis.

Diurnal Effects: Surface based warm core features during night-time cooling would decrease in intensity as the atmosphere affecting the feature would become relatively colder than during the hottest part of the day resulting in cyclolysis.

Practice: Prognosis Rules / Forecast Techniques

Climatology

Climatology is the study of past trends as well as a record of historic weather observations. This information can be accessed through the 14th Weather Squadron or FNMOC. The information that climatology provides can be an invaluable tool when developing your forecast.

14 Weather Squadron (14WS)
The 14WS provides historical weather data for forecasters to search and tailor data to create user-defined special interest areas. The website uses latitude and longitude coordinates or a weather stations ICAO.

Operational Climatic Data Summaries (OCDS)

A quick reference listing most meteorological elements for a single location are Operational Climatic Data Summaries (OCDS). This will show you the maximum, average, and minimum values for dozens of weather parameters that can help you determine if a forecast is reasonable or possible.

Temperature

PARAMETER	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANN
Extreme Max (F)	81	81	90	93	97	102	102	106	99	95	88	81	106
Average Max (F)	61	65	71	76	83	88	89	89	87	81	70	65	77
Daily Average (F)	55	59	64	69	76	80	81	81	79	72	62	59	70
Average Min (F)	45	50	56	61	68	75	76	76	73	64	52	50	62
Extreme Min (F)	10	15	24	36	45	57	60	61	45	34	25	12	10
Max Diurnal Range (F)	40	36	38	31	31	25	29	27	29	32	38	34	40
Average # Days >= 90 F	0	0	0	0	2	9	16	17	11	1	0	0	55
Average # Days <= 32 F	4	1	#	0	0	0	0	0	0	0	1	1	7
Average # Days <= 0 F	0	0	0	0	0	0	0	0	0	0	0	0	0

* = Data not Available # = Occurrences rounded to zero

Wind

PARAMETER	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANN
Prevailing Direction	$N	S	S	S	S	S	$SW	$SW	NE	$NE	$NNE	$NNE	$S
Average Prevailing Speed (kts)	9.1	7.8	7.8	8.5	8.5	9.5	8.9	9.7	6.6	6.0	7.6	7.2	8.4
Extreme Max Speed (kts)	29.9	29.9	35.0	42.0	29.9	35.0	31.1	69.9	64.9	55.9	28.9	28.9	69.9
Extreme Max Gust (kts)	60.0	54.0	60.0	61.0	68.0	58.1	56.9	112.1	85.1	78.1	48.0	55.9	112.1
Average Speed (kts)	6.6	7.0	7.2	7.4	7.2	7.2	5.2	6.8	5.8	6.6	5.8	6.2	6.6

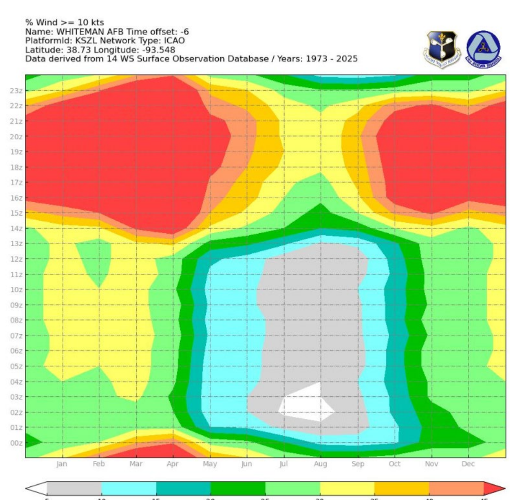
* = Data not Available # = Occurrences rounded to zero $ = Percent of Calm > Prevailing Direction Std Dev = Standard Deviation
*** = Variable Wind Direction Predominant

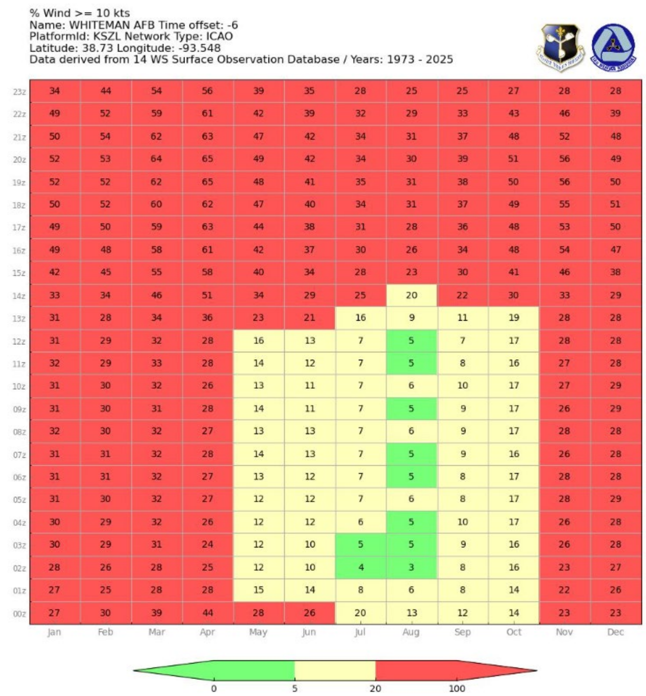
MODCURVES

Another useful climatology product is MODCURVES. They display diurnal changes of temperature and pressure which can be an effective forecast tool when the air mass does not change during the forecast period.

Ceiling / Visibility / Wind Probability

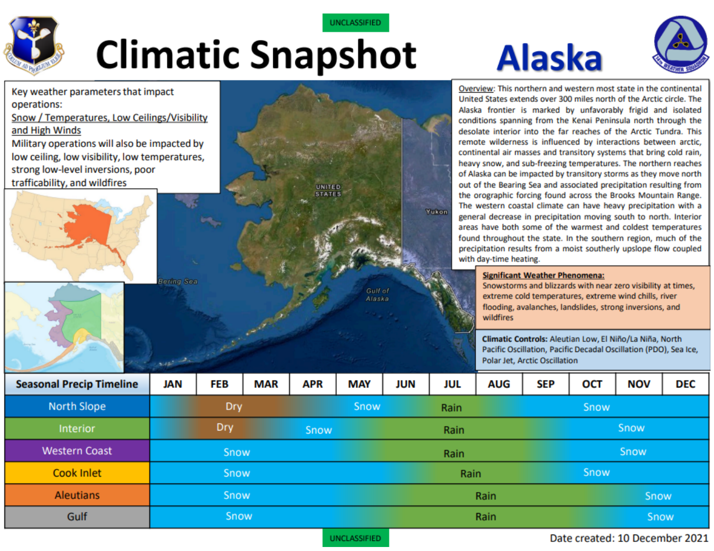
The Ceiling/Visibility/Wind products is used to highlight seasonal and monthly trends and thresholds of interest.





Climate Snapshot

There are multiple climatological products available throughout the 14WS. For locations that do not have climatology data available, Climate Snapshots are available to obtain a brief narrative on the weather conditions for individual countries.



FNMOC Climo Support Pages

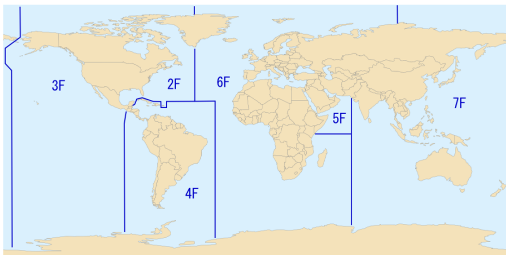
Climatology data for oceanic data can be accessed through FNMOC. The user can select a fleet that focuses on the area of interest. Here you can further focus on the region of information and view climatology data for several parameters broken down by monthly averages.

Fleet Numerical Meteorology and Oceanography Center

Home of the Climate Support Pages (CSPs)

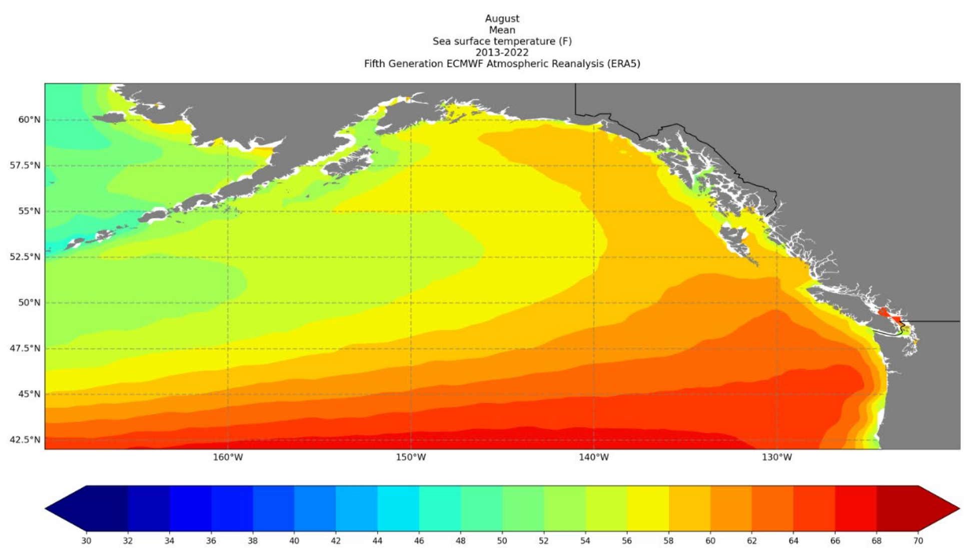
Select a Fleet:

- Second Fleet
- Third Fleet
- Fourth Fleet
- Fifth Fleet
- Sixth Fleet
- Seventh Fleet



Regional Climatology Support Products

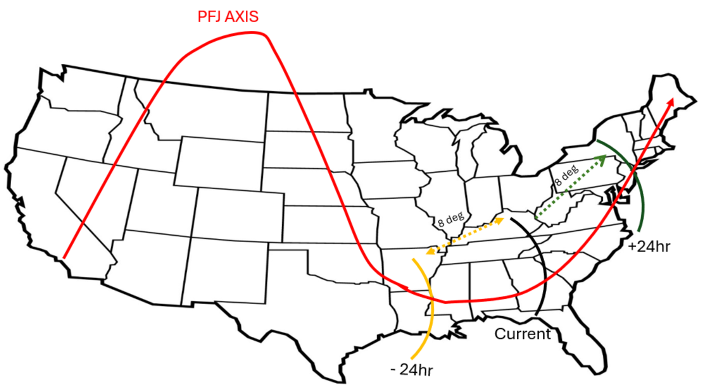
Parameter	Pacific Transit	Sea of Okhotsk	Bering Sea	Gulf of Alaska	Pacific Northwest	Southern California	Hawaiian Islands	Western Central America
2-meter Air Temperature [degrees Fahrenheit]	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Maximum 2-m Air Temperature [degrees Fahrenheit]	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Minimum 2-m Air Temperature [degrees Fahrenheit]	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Relative Humidity [%]	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Surface Sea Temperatures [degrees Fahrenheit]	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Currents	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Cloud Climatology								
Cloud Base Height: Frequency of Occurence Below 3000 ft	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Cloud Base Height: Frequency of Occurence Below 1000 ft	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec	Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec



Practice: Navigating the 14^{th} WS / FNMOC

Continuity & Extrapolation

Continuity is a record of past positions of a specific weather feature. It shows movement of the feature with respect to time. Extrapolation uses a feature's past positions to determine a "first guess" at locating its future position. For example, if a major-short wave trough has been moving 8° per day for the last two days, extrapolation assumes its position will be 8° downstream from its current position in 24 hours.



Practice: Continuity & Extrapolation

PC: Forecast Tools